ACA Dependent-Coverage Mandate: Replication

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Paper

Akosa Antwi, Yaa, Asako S. Moriya, and Kosali Simon. 2013. "Effects of Federal Policy to Insure Young Adults: Evidence from the 2010 Affordable Care Act's Dependent-Coverage Mandate." *American Economic Journal: Economic Policy* 5(4): 1–28.

Replication target

Table 1 (summary statistics: demographic, socioeconomic and insurance characteristics) and **Table 2** (main DD results: effect of the ACA on coverage of young adults aged 19–25).

Research design

Difference-in-differences around the age-26 cutoff. Ages 19–25 (eligible, `fedelig=1`) are compared with ages 16–18 and 27–29, before and after the mandate took effect. The DD estimate is the coefficient on `elig_oct10` (eligible × after Oct 2010).

Context

Completed as a lab for Prof. Thakral's MPA 2040 course at Brown University. Code: mcallister_aca_dd_estimate.do.

Part 1: Code

CODE	WHAT IT DOES

`clear all`
`cd "/Users/alexander/Documents/phd/stata/"`
`log using lab3.log, replace`
`use lab3.dta, clear`
	Setup: clear memory, set the folder, start a log file, load the data. **p_weight** is the survey weight.

`display "Table 1. Demographic, Socioeconomic and`
`Insurance Characteristics"`
 	
`* all observations`
`sum anyhi emphi_dep emphi indiv govhi employed`
`unemployed age white black hispanic mar ft_student`
`hsdo hsg somcol colgrd bad_hlth [weight=p_weight]`
	Table 1, first column. Weighted means and standard deviations. `[weight=p_weight]` makes the means represent the population. Insurance outcomes: anyhi (any), emphi_dep (parent's employer), emphi (own employer), indiv (individual purchase), govhi (government).

`* by age group` `forvalues k = 1 / 3 {` `sum anyhi emphi_dep ... bad_hlth [weight=p_weight]` `if age_group==`k'` `}`	**Table 1, other columns.** A loop: runs the same summary once for each age_group, where `k' is replaced by 1, 2, 3. Group 1 = ages 16–18, 2 = ages 19–25 (eligible), 3 = ages 27–29.
`local effects mar_sep10 after_oct10 fedelig` `elig_mar10 elig_oct10`	**Local macro.** Saves a list of variable names under the name *effects*, so regressions can reuse it as `effects'. These are the policy variables: period dummies, the eligible-group dummy fedelig, and its interactions with the periods. **elig_oct10 is the DD estimate.**

`local exp_vars age17-age18 age20-age25 age27-age29`
`trend female ue hispanic white asian other student`
`mar fpl_ratio fpl_ratio_2 ue_treat y2009 y2010`
`month2-month12 st_trend_1-st_trend_55`
	Control variables stored in a second macro: single-year age dummies, time trend, demographics, poverty ratio (and square), unemployment, year and month dummies, state-specific trends. `a-b` means every variable from a to b.

`display "Table 2: ... Main DD Results (regression)"` `foreach instype in anyhi emphi_dep indiv emphi govhi {` `areg `instype' `effects' `exp_vars' [weight=p_weight],` `absorb(fipstate) cluster(fipstate)` `}`	**Table 2, regression DD.** Loops over the five coverage outcomes, one regression each. `areg` is OLS with state fixed effects: `absorb(fipstate)` includes a dummy for each state without printing them. `cluster(fipstate)` allows errors to be correlated within a state. Read the **elig_oct10** coefficient.

`display "Table 2: ... Main DD Results (unconditional DD)"`
`sum anyhi emphi_dep indiv emphi govhi [weight=p_weight]`
`if after_oct10==0 & mar_sep10==0 & fedelig==1`
`sum ... if after_oct10==0 & mar_sep10==0 & fedelig==0`
`sum ... if after_oct10==1 & fedelig==1`
`sum ... if after_oct10==1 & fedelig==0`
	Table 2, unconditional DD. Four weighted means, one per cell of the 2×2 table: eligible before, comparison before, eligible after, comparison after. DD = (eligible after – before) – (comparison after – before), with no controls.

`log close`	Stops the log.

Part 2: Output behind Table 1 (summary statistics)

STATA OUTPUT (trimmed)

```
Variable    |    Obs      Mean  Std.dev.
-------------------------------------
anyhi       | 150,997  .7266878  .4456612
emphi_dep   | 150,997  .2672525  .4425267
emphi       | 150,997  .1981379  .3985979
indiv       | 150,997  .0279155  .1647313
govhi       | 150,997  .1577302  .3644891
employed    | 150,997  .5692072  .4951888
unemployed  | 150,997  .0796091  .2706879
age         | 150,997  22.22218  4.090602
white       | 150,997  .6020064  .4894857
black       | 150,997  .1358809  .3426632
hispanic    | 150,997   .189563  .3919565
mar         | 150,997  .1796552  .3839013
ft_student  | 150,997  .4192515  .4934382
hsdo        | 150,997  .2572088  .4370969
hsg         | 150,997  .2715724   .444772
somcol      | 150,997  .3331745  .4713499
colgrd      | 150,997  .1127398  .3162755
bad_hlth    | 125,791  .5847029  .4927751
```

```
Variable    |    Obs      Mean  Std.dev.
-------------------------------------
anyhi       |  39,886  .8696125  .3367335
emphi_dep   |  39,886  .5228381  .4994844
emphi       |  39,886  .0240132   .153092
indiv       |  39,886  .0105617  .1022274
govhi       |  39,886  .2559854  .4364191
employed    |  39,886  .2528022   .434624
unemployed  |  39,886  .0631775  .2432849
age         |  39,886  17.01745  .8170374
white       |  39,886  .5749358  .4943589
black       |  39,886  .1460453   .353156
hispanic    |  39,886  .1995477  .3996654
mar         |  39,886  .0125882  .1114902
ft_student  |  39,886  .8850394  .3189784
hsdo        |  39,886  .7702667  .4206666
hsg         |  39,886  .1779084  .3824404
somcol      |  39,886  .0511637  .2203343
colgrd      |  39,886  .0005539  .0235278
bad_hlth    |  34,604  .4863484  .4998208
```

NOTES

All observations (ages 16–29)

Table 1, all-observations column. Obs and Mean are shown. The log also prints weight, min and max (omitted here).

age_group == 1: ages 16–18 (comparison)

Table 1, ages 16–18. Most are covered through a parent (52%); few have own-employer coverage (2%).

STATA OUTPUT (trimmed)

NOTES

```
Variable    |     Obs      Mean  Std.dev.
-------------------------------------------
anyhi       |  78,212  .6780652  .4672211
emphi_dep   |  78,212  .2638311  .4407116
emphi       |  78,212  .1891256  .3916108
indiv       |  78,212  .0320453  .1761216
govhi       |  78,212  .1270493  .3330304
employed    |  78,212  .6278902  .4833705
unemployed  |  78,212  .0886333  .2842155
age         |  78,212  21.96307    2.0038
white       |  78,212  .6131207  .4870387
black       |  78,212  .1354633  .3422199
hispanic    |  78,212  .1827836  .3864915
mar         |  78,212  .1403913  .3473947
ft_student  |  78,212  .3613706  .4804007
hsdo        |  78,212  .0978671  .2971367
hsg         |  78,212  .3248399   .468318
somcol      |  78,212  .4522739  .4977202
colgrd      |  78,212  .1134127  .3170986
bad_hlth    |  63,708  .5958978  .4907213
```

age_group == 2: ages 19–25 (eligible)

Table 1, ages 19–25. The treated group. Parent coverage falls to 26% as people age out of their parents' plans; any coverage is 68%.

```
Variable    |     Obs      Mean  Std.dev.
-------------------------------------------
anyhi       |  32,899  .6932211  .4611638
emphi_dep   |  32,899  .0192395  .1373677
emphi       |  32,899  .3926567  .4883489
indiv       |  32,899  .0359751  .1862308
govhi       |  32,899  .1285387  .3346937
employed    |  32,899  .7535825   .430931
unemployed  |  32,899  .0757203  .2645541
age         |  32,899  28.01334  .8148771
white       |  32,899  .6040539  .4890604
black       |  32,899  .1266519  .3325877
hispanic    |  32,899  .1948453  .3960876
mar         |  32,899   .435261  .4957987
ft_student  |  32,899  .0836075  .2768025
hsdo        |  32,899  .1028329  .3037451
hsg         |  32,899  .2452888  .4302648
somcol      |  32,899  .3470363  .4760347
colgrd      |  32,899  .2234691  .4165765
bad_hlth    |  27,479  .6627204  .4727898
```

age_group == 3: ages 27–29 (comparison)

Table 1, ages 27–29. Parent coverage is nearly gone (2%); own-employer coverage is 39%.

Part 2: Output behind Table 2 (main DD results)

STATA OUTPUT (policy rows only)

```
Outcome: anyhi
            |   Coef.    Std.err.      t    P>|t|
-------------------------------------------------
mar_sep10   |-.0079679  .0065126    -1.22   0.227
after_oct10 |-.0242815  .0065774    -3.69   0.001
fedelig     |-.0820124  .0203237    -4.04   0.000
elig_mar10  |-.0017525  .0063481    -0.28   0.784
elig_oct10  | .0317839  .0074088     4.29   0.000
N = 150,997   R-squared = 0.1700

Outcome: emphi_dep
            |   Coef.    Std.err.      t    P>|t|
-------------------------------------------------
mar_sep10   |-.0266264  .0072416    -3.68   0.001
after_oct10 | -.045641  .0074914    -6.09   0.000
fedelig     |-.0543472  .0200543    -2.71   0.009
elig_mar10  | .0238501  .0055182     4.32   0.000
elig_oct10  |  .070174  .0069384    10.11   0.000
N = 150,997   R-squared = 0.3344

Outcome: indiv
            |   Coef.    Std.err.      t    P>|t|
-------------------------------------------------
mar_sep10   |-.0024703  .0029121    -0.85   0.400
after_oct10 |-.0021448  .0045187    -0.47   0.637
fedelig     | .0178183  .0091409     1.95   0.057
elig_mar10  | .0024879   .003109     0.80   0.427
elig_oct10  |-.0079611  .0023347    -3.41   0.001
N = 150,997   R-squared = 0.0192

Outcome: emphi
            |   Coef.    Std.err.      t    P>|t|
-------------------------------------------------
mar_sep10   | .0063079  .0061318     1.03   0.309
after_oct10 |   .01438  .0090686     1.59   0.119
fedelig     | .0302303   .024434     1.24   0.222
elig_mar10  | -.017271  .0053055    -3.26   0.002
elig_oct10  |-.0311837   .005992    -5.20   0.000
N = 150,997   R-squared = 0.1865

Outcome: govhi
            |   Coef.    Std.err.      t    P>|t|
-------------------------------------------------
mar_sep10   | .0154551  .0069313     2.23   0.030
after_oct10 | .0117439  .0081755     1.44   0.157
fedelig     |-.0785978  .0202596    -3.88   0.000
elig_mar10  |-.0105588  .0053609    -1.97   0.054
elig_oct10  |-.0024601  .0057499    -0.43   0.671
N = 150,997   R-squared = 0.1245
```

NOTES

Regression DD. Only the policy rows are shown; the ~100 control rows (ages, demographics, trends) are left out. Standard errors are clustered by state (51 clusters).

Any coverage. DD estimate (elig_oct10): +3.2 percentage points (p = 0.000).

Coverage through a parent (dependent). DD estimate (elig_oct10): +7.0 percentage points (p = 0.000).

Individually purchased. DD estimate (elig_oct10): -0.8 percentage points (p = 0.001).

Coverage through own employer. DD estimate (elig_oct10): -3.1 percentage points (p = 0.000).

Government coverage. DD estimate (elig_oct10): -0.2 percentage points (p = 0.671).

Part 2: Unconditional DD (no controls)

STATA OUTPUT (trimmed)

```
Variable    |    Obs      Mean  Std.dev.
-------------------------------------
anyhi       |  39,667  .6646935  .4721035
emphi_dep   |  39,667  .2338893  .4233079
indiv       |  39,667  .0348704  .1834538
emphi       |  39,667  .2040713  .4030264
govhi       |  39,667  .1231835  .3286519
```

Cell 1: Eligible (19–25), BEFORE.

```
Variable    |    Obs      Mean  Std.dev.
-------------------------------------
anyhi       |  37,521  .7811376  .4134806
emphi_dep   |  37,521  .2800574   .449033
indiv       |  37,521  .0234069  .1511939
emphi       |  37,521  .2075061  .4055264
govhi       |  37,521   .181974  .3858283
```

Cell 2: Comparison, BEFORE.

```
Variable    |    Obs      Mean  Std.dev.
-------------------------------------
anyhi       |  25,101   .702194  .4573029
emphi_dep   |  25,101  .3069489  .4612372
indiv       |  25,101   .025985  .1590936
emphi       |  25,101  .1712954  .3767744
govhi       |  25,101  .1326932  .3392496
```

Cell 3: Eligible (19–25), AFTER Oct 2010.

```
Variable    |    Obs      Mean  Std.dev.
-------------------------------------
anyhi       |  22,838  .7826939  .4124216
emphi_dep   |  22,838  .2628801  .4402075
indiv       |  22,838  .0230896  .1501915
emphi       |  22,838  .2096526  .4070696
govhi       |  22,838  .2004835  .4003709
```

Cell 4: Comparison, AFTER Oct 2010.

```
           |  Elig.chg  Comp.chg     DD
-----------------------------------------
anyhi      |     +3.8      +0.2    +3.6
emphi_dep  |     +7.3      -1.7    +9.0
indiv      |     -0.9      -0.0    -0.9
emphi      |     -3.3      +0.2    -3.5
govhi      |     +1.0      +1.9    -0.9
(percentage points)
```

Unconditional DD, computed from the four cells. For coverage through a parent: eligible rises 7.3 pp, comparison falls 1.7 pp, so DD = +9.0 pp. The regression DD with controls (+7.0 pp) is smaller.